LOGAN YUAN

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PROFESSIONAL SUMMARY

Software and Biomedical Engineering graduate from McMaster University with hands-on experience building data pipelines, motion analysis systems, and full-stack applications. Proficient in Python, cloud platforms (Azure, AWS), and machine learning frameworks. Passionate about applying engineering principles to real-world problems in healthcare and technology, with a track record of delivering functional prototypes and tools that reduce manual effort.

EDUCATION

Bachelor of Software and Biomedical Engineering — Sept 2020 – Jun 2025
McMaster University, Hamilton, Ontario

Capstone Project: **Virtual Dragon Boat Coach** — Sept 2024 – Apr 2025
github.com/JamesParkGH/Virtual_Dragon_Boat_Coach

- Built a motion analysis pipeline in Python that consumes 60 Frames Per Second data from the OpenCap API to evaluate dragon boat athletes' stroke technique and overall form.
- Implemented evaluation algorithms in Python to extract stroke count, joint positions, and joint angles from OpenCap datasets.
- Achieved over 90% accuracy in identifying problematic technique across recorded training sessions.
- Designed a lightweight web interface using Python and HTML to let athletes upload sessions and visualize stroke metrics and feedback, reducing the need for manual video review by coaches.

Course Project: **Neck Pain (NSNP) Monitor** — Dec 2022 – Apr 2023

- Prototyped a real-time neck posture monitoring device using an Arduino micro-controller and an orientation sensor to track users' neck angle during everyday office computer use.
- Programmed the micro-controller in C to stream continuous sensor readings via the Arduino serial communication interface to a Python backend at 60 Hz, enabling low-latency monitoring of user posture.
- Designed a Python algorithm that analyzes time-series neck-angle data to identify postures exceeding ergonomic risk thresholds, demonstrating effective performance across all three test users.

SKILLS

Programming Languages	Python, Java, JavaScript, TypeScript, SQL, HTML
Technical Skills	Git, Azure, AWS, Linux, PyTorch, MATLAB, Flask, React
Certifications	Microsoft Certified: Azure AI Fundamentals (AI-900)

PERSONAL PROJECTS

Stock Tracker — Dec 2025 – Present
github.com/yuanq10/StockTracker

- Created a Python-based stock monitoring tool that delivers automated notifications when configurable conditions are triggered, reducing the time required for daily market analysis.
- Developed a data pipeline that automatically retrieves and processes stock market data using the yfinance API, ensuring the monitoring system stays up to date without requiring users to manually collect market data.
- Designed a user-friendly GUI using CustomTkinter, applying human-computer interaction principles to enable efficient watchlist management, indicator configuration, and data analysis.